FLOW MATCHING FOR GENERATIVE MODELING

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ABSTRACT

We introduce a new paradigm for generative modeling built on Continuous Normalizing Flows (CNFs), allowing us to train CNFs at unprecedented scale. Specifically, we present the notion of Flow Matching (FM), a simulation-free approach for training CNFs based on regressing vector fields of fixed conditional probability paths. Flow Matching is compatible with a general family of Gaussian probability paths for transforming between noise and data samples—which subsumes existing diffusion paths as specific instances. Interestingly, we find that employing FM with diffusion paths results in a more robust and stable alternative for training diffusion models. Furthermore, Flow Matching opens the door to training CNFs with other, non-diffusion probability paths. An instance of particular interest is using Optimal Transport (OT) displacement interpolation to define the conditional probability paths. These paths are more efficient than diffusion paths, provide faster training and sampling, and result in better generalization. Training CNFs using Flow Matching on ImageNet leads to consistently better performance than alternative diffusion-based methods in terms of both likelihood and sample quality, and allows fast and reliable sample generation using off-the-shelf numerical ODE solvers.

1 INTRODUCTION

Deep generative models are a class of deep learning algorithms aimed at estimating and sampling from an unknown data distribution. The recent influx of amazing advances in generative modeling, *e.g.*, for image generation Ramesh et al. (2022); Rombach et al. (2022), is mostly facilitated by the scalable and relatively stable training of diffusion-based models Ho et al. (2020); Song et al. (2020b). However, the restriction to simple diffusion processes leads to a rather confined space of sampling probability paths, resulting in very long training times and the need to adopt specialized methods (*e.g.*, Song et al. (2020a); Zhang & Chen (2022)) for efficient sampling.

In this work we consider the general and deterministic framework of Continuous Normalizing Flows (CNFs; Chen et al. (2018)). CNFs are capable of modeling arbitrary probability path and are in particular known to encompass the probability paths modeled by diffusion processes (Song et al., 2021). However, aside from diffusion that can be trained efficiently via, *e.g.*, denoising score matching (Vincent, 2011), no scalable CNF training algorithms are known. Indeed, maximum likelihood training (*e.g.*, Grathwohl et al. (2018)) require expensive numerical ODE simulations, while existing simulation-free methods either involve intractable integrals (Rozen et al., 2021) or biased gradients (Ben-Hamu et al., 2022).

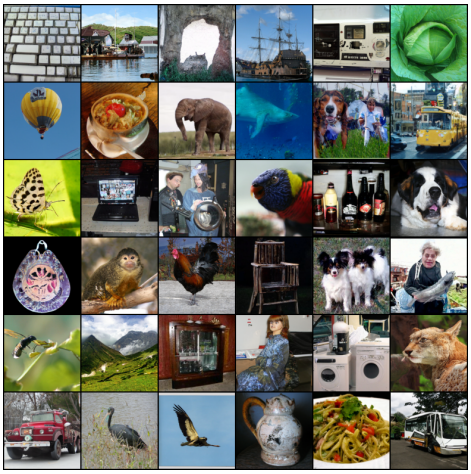


Figure 1: Unconditional ImageNet-128 samples of a CNF trained using Flow Matching with Optimal Transport probability paths.

The goal of this work is to propose Flow Matching (FM), an efficient simulation-free approach to training CNF models, allowing the adoption of general probability paths to supervise CNF training. Importantly, FM breaks the barriers for scalable CNF training beyond diffusion, and sidesteps the need to reason about diffusion processes to directly work with probability paths.

In particular, we propose the Flow Matching objective (Section 3), a simple and intuitive training objective to regress onto a target vector field that generates a desired probability path. We first show that we can construct such target vector fields through per-example (*i.e.*, conditional) formulations. Then, inspired by denoising score matching, we show that a per-example training objective, termed Conditional Flow Matching (CFM), provides equivalent gradients and does not require explicit knowledge of the intractable target vector field. Furthermore, we discuss a general family of per-example probability paths (Section 4) that can be used for Flow Matching, which subsumes existing diffusion paths as special instances. Even on diffusion paths, we find that using FM provides more robust and stable training, and achieves superior performance compared to score matching. Furthermore, this family of probability paths also includes a particularly interesting case: the vector field that corresponds to an Optimal Transport (OT) displacement interpolant (McCann, 1997). We find that conditional OT paths are simpler than diffusion paths, forming straight line trajectories whereas diffusion paths result in curved paths. These properties seem to empirically translate to faster training, faster generation, and better performance.

We empirically validate Flow Matching and the construction via Optimal Transport paths on ImageNet, a large and highly diverse image dataset. We find that we can easily train models to achieve favorable performance in both likelihood estimation and sample quality amongst competing diffusion-based methods. Furthermore, we find that our models produce better trade-offs between computational cost and sample quality compared to prior methods. Figure 1 depicts selected unconditional ImageNet 128×128 samples from our model.

2 Preliminaries: Continuous Normalizing Flows

Let $\mathbb{R}^d$ denote the data space with data points $x = (x^1, \ldots, x^d) \in \mathbb{R}^d$. Two important objects we use in this paper are: the *probability density path* $p : [0,1] \times \mathbb{R}^d \to \mathbb{R}_{>0}$, which is a time dependent[1] probability density function, *i.e.*, $\int p_t(x)dx = 1$, and a *time-dependent vector field*, $v : [0,1] \times \mathbb{R}^d \to \mathbb{R}^d$. A vector field v_t can be used to construct a time-dependent diffeomorphic map, called a *flow*, $\phi : [0,1] \times \mathbb{R}^d \to \mathbb{R}^d$, defined via the ordinary differential equation (ODE):

$$\frac{d}{dt}\phi_t(x) = v_t(\phi_t(x)) \tag{1}$$

$$\phi_0(x) = x \tag{2}$$

Previously, Chen et al. (2018) suggested modeling the vector field v_t with a neural network, $v_t(x;\theta)$, where $\theta \in \mathbb{R}^p$ are its learnable parameters, which in turn leads to a deep parametric model of the flow ϕ_t, called a *Continuous Normalizing Flow* (CNF). A CNF is used to reshape a simple prior density p_0 (*e.g.*, pure noise) to a more complicated one, p_1, via the push-forward equation

$$p_t = [\phi_t]_* p_0 \tag{3}$$

where the push-forward (or change of variables) operator $*$ is defined by

$$[\phi_t]_* p_0(x) = p_0(\phi_t^{-1}(x)) \det \left[\frac{\partial \phi_t^{-1}}{\partial x}(x) \right]. \tag{4}$$

A vector field v_t is said to *generate* a probability density path p_t if its flow ϕ_t satisfies equation 3. One practical way to test if a vector field generates a probability path is using the continuity equation, which is a key component in our proofs, see Appendix B. We recap more information on CNFs, in particular how to compute the probability $p_1(x)$ at an arbitrary point $x \in \mathbb{R}^d$ in Appendix C.

3 Flow Matching

Let x_1 denote a random variable distributed according to some unknown data distribution $q(x_1)$. We assume we only have access to data samples from $q(x_1)$ but have no access to the density function itself. Furthermore, we let p_t be a probability path such that $p_0 = p$ is a simple distribution, *e.g.*, the standard normal distribution $p(x) = \mathcal{N}(x|0, I)$, and let p_1 be approximately equal in distribution to q. We will later discuss how to construct such a path. The Flow Matching objective is then designed to match this target probability path, which will allow us to flow from p_0 to p_1.

[1] We use subscript to denote the time parameter, *e.g.*, $p_t(x)$.

Given a target probability density path $p_t(x)$ and a corresponding vector field $u_t(x)$, which generates $p_t(x)$, we define the Flow Matching (FM) objective as

$$\mathcal{L}_{\text{FM}}(\theta) = \mathbb{E}_{t,p_t(x)}\|v_t(x) - u_t(x)\|^2, \tag{5}$$

where θ denotes the learnable parameters of the CNF vector field v_t (as defined in Section 2), $t \sim \mathcal{U}[0,1]$ (uniform distribution), and $x \sim p_t(x)$. Simply put, the FM loss regresses the vector field u_t with a neural network v_t. Upon reaching zero loss, the learned CNF model will generate $p_t(x)$.

Flow Matching is a simple and attractive objective, but naïvely on its own, it is intractable to use in practice since we have no prior knowledge for what an appropriate p_t and u_t are. There are many choices of probability paths that can satisfy $p_1(x) \approx q(x)$, and more importantly, we generally don't have access to a closed form u_t that generates the desired p_t. In this section, we show that we can construct both p_t and u_t using probability paths and vector fields that are only defined *per sample*, and an appropriate method of aggregation provides the desired p_t and u_t. Furthermore, this construction allows us to create a much more tractable objective for Flow Matching.

3.1 Constructing p_t, u_t from conditional probability paths and vector fields

A simple way to construct a target probability path is via a mixture of simpler probability paths: Given a particular data sample x_1 we denote by $p_t(x|x_1)$ a *conditional probability path* such that it satisfies $p_0(x|x_1) = p(x)$ at time $t = 0$, and we design $p_1(x|x_1)$ at $t = 1$ to be a distribution concentrated around $x = x_1$, *e.g.*, $p_1(x|x_1) = \mathcal{N}(x|x_1, \sigma^2 I)$, a normal distribution with x_1 mean and a sufficiently small standard deviation $\sigma > 0$. Marginalizing the conditional probability paths over $q(x_1)$ give rise to *the marginal probability path*

$$p_t(x) = \int p_t(x|x_1)q(x_1)dx_1, \tag{6}$$

where in particular at time $t = 1$, the marginal probability p_1 is a mixture distribution that closely approximates the data distribution q,

$$p_1(x) = \int p_1(x|x_1)q(x_1)dx_1 \approx q(x). \tag{7}$$

Interestingly, we can also define a *marginal vector field*, by "marginalizing" over the conditional vector fields in the following sense (we assume $p_t(x) > 0$ for all t and x):

$$u_t(x) = \int u_t(x|x_1)\frac{p_t(x|x_1)q(x_1)}{p_t(x)}dx_1, \tag{8}$$

where $u_t(\cdot|x_1) : \mathbb{R}^d \to \mathbb{R}^d$ is a conditional vector field that generates $p_t(\cdot|x_1)$. It may not seem apparent, but this way of aggregating the conditional vector fields actually results in the correct vector field for modeling the marginal probability path.

Our first key observation is this:

The marginal vector field (equation 8) generates the marginal probability path (equation 6).

This provides a surprising connection between the conditional VFs (those that generate conditional probability paths) and the marginal VF (those that generate the marginal probability path). This connection allows us to break down the unknown and intractable marginal VF into simpler conditional VFs, which are much simpler to define as these only depend on a single data sample. We formalize this in the following theorem.

Theorem 1. *Given vector fields $u_t(x|x_1)$ that generate conditional probability paths $p_t(x|x_1)$, for any distribution $q(x_1)$, the marginal vector field u_t in equation 8 generates the marginal probability path p_t in equation 6, i.e., u_t and p_t satisfy the continuity equation (equation 26).*

The full proofs for our theorems are all provided in Appendix A. Theorem 1 can also be derived from the Diffusion Mixture Representation Theorem in Peluchetti (2021) that provides a formula for the marginal drift and diffusion coefficients in diffusion SDEs.

3.2 Conditional Flow Matching

Unfortunately, due to the intractable integrals in the definitions of the marginal probability path and VF (equations 6 and 8), it is still intractable to compute u_t, and consequently, intractable to naïvely compute an unbiased estimator of the original Flow Matching objective. Instead, we propose a simpler objective, which surprisingly will result in the same optima as the original objective. Specifically, we consider the *Conditional Flow Matching* (CFM) objective,

$$\mathcal{L}_{\text{CFM}}(\theta) = \mathbb{E}_{t,q(x_1),p_t(x|x_1)} \left\| v_t(x) - u_t(x|x_1) \right\|^2, \tag{9}$$

where $t \sim \mathcal{U}[0,1]$, $x_1 \sim q(x_1)$, and now $x \sim p_t(x|x_1)$. Unlike the FM objective, the CFM objective allows us to easily sample unbiased estimates as long as we can efficiently sample from $p_t(x|x_1)$ and compute $u_t(x|x_1)$, both of which can be easily done as they are defined on a per-sample basis. Our second key observation is therefore:

The FM (equation 5) and CFM (equation 9) objectives have identical gradients w.r.t. θ.

That is, optimizing the CFM objective is equivalent (in expectation) to optimizing the FM objective. Consequently, this allows us to train a CNF to generate the marginal probability path p_t—which in particular, approximates the unknown data distribution q at t=1— without ever needing access to either the marginal probability path or the marginal vector field. We simply need to design suitable *conditional* probability paths and vector fields. We formalize this property in the following theorem.

Theorem 2. *Assuming that $p_t(x) > 0$ for all $x \in \mathbb{R}^d$ and $t \in [0,1]$, then, up to a constant independent of θ, $\mathcal{L}_{CFM}$ and $\mathcal{L}_{FM}$ are equal. Hence, $\nabla_\theta \mathcal{L}_{FM}(\theta) = \nabla_\theta \mathcal{L}_{CFM}(\theta)$.*

4 Conditional Probability Paths and Vector Fields

The Conditional Flow Matching objective works with any choice of conditional probability path and conditional vector fields. In this section, we discuss the construction of $p_t(x|x_1)$ and $u_t(x|x_1)$ for a general family of Gaussian conditional probability paths. Namely, we consider conditional probability paths of the form

$$p_t(x|x_1) = \mathcal{N}(x \mid \mu_t(x_1), \sigma_t(x_1)^2 I), \tag{10}$$

where $\mu : [0,1] \times \mathbb{R}^d \to \mathbb{R}^d$ is the time-dependent mean of the Gaussian distribution, while $\sigma : [0,1] \times \mathbb{R} \to \mathbb{R}_{>0}$ describes a time-dependent scalar standard deviation (std). We set $\mu_0(x_1) = 0$ and $\sigma_0(x_1) = 1$, so that all conditional probability paths converge to the same standard Gaussian noise distribution at $t = 0$, $p(x) = \mathcal{N}(x|0, I)$. We then set $\mu_1(x_1) = x_1$ and $\sigma_1(x_1) = \sigma_{\text{min}}$, which is set sufficiently small so that $p_1(x|x_1)$ is a concentrated Gaussian distribution centered at x_1.

There is an infinite number of vector fields that generate any particular probability path (*e.g.*, by adding a divergence free component to the continuity equation, see equation 26), but the vast majority of these is due to the presence of components that leave the underlying distribution invariant—for instance, rotational components when the distribution is rotation-invariant—leading to unnecessary extra compute. We decide to use the simplest vector field corresponding to a canonical transformation for Gaussian distributions. Specifically, consider the flow (conditioned on x_1)

$$\psi_t(x) = \sigma_t(x_1)x + \mu_t(x_1). \tag{11}$$

When x is distributed as a standard Gaussian, $\psi_t(x)$ is the affine transformation that maps to a normally-distributed random variable with mean $\mu_t(x_1)$ and std $\sigma_t(x_1)$. That is to say, according to equation 4, ψ_t pushes the noise distribution $p_0(x|x_1) = p(x)$ to $p_t(x|x_1)$, *i.e.*,

$$[\psi_t]_* \, p(x) = p_t(x|x_1). \tag{12}$$

This flow then provides a vector field that generates the conditional probability path:

$$\frac{d}{dt}\psi_t(x) = u_t(\psi_t(x)|x_1). \tag{13}$$

Reparameterizing $p_t(x|x_1)$ in terms of just x_0 and plugging equation 13 in the CFM loss we get

$$\mathcal{L}_{\text{CFM}}(\theta) = \mathbb{E}_{t,q(x_1),p(x_0)} \left\| v_t(\psi_t(x_0)) - \frac{d}{dt}\psi_t(x_0) \right\|^2. \tag{14}$$

Since ψ_t is a simple (invertible) affine map we can use equation 13 to solve for u_t in a closed form. Let f' denote the derivative with respect to time, *i.e.*, $f' = \frac{d}{dt} f$, for a time-dependent function f.

Theorem 3. *Let $p_t(x|x_1)$ be a Gaussian probability path as in equation 10, and ψ_t its corresponding flow map as in equation 11. Then, the unique vector field that defines ψ_t has the form:*

$$u_t(x|x_1) = \frac{\sigma_t'(x_1)}{\sigma_t(x_1)}\left(x - \mu_t(x_1)\right) + \mu_t'(x_1). \tag{15}$$

Consequently, $u_t(x|x_1)$ generates the Gaussian path $p_t(x|x_1)$.

4.1 Special instances of Gaussian conditional probability paths

Our formulation is fully general for arbitrary functions $\mu_t(x_1)$ and $\sigma_t(x_1)$, and we can set them to any differentiable function satisfying the desired boundary conditions. We first discuss the special cases that recover probability paths corresponding to previously-used diffusion processes. Since we directly work with probability paths, we can simply depart from reasoning about diffusion processes altogether. Therefore, in the second example below, we directly formulate a probability path based on the Wasserstein-2 optimal transport solution as an interesting instance.

Example I: Diffusion conditional VFs. Diffusion models start with data points and gradually add noise until it approximates pure noise. These can be formulated as stochastic processes, which have strict requirements in order to obtain closed form representation at arbitrary times t, resulting in Gaussian conditional probability paths $p_t(x|x_1)$ with specific choices of mean $\mu_t(x_1)$ and std $\sigma_t(x_1)$ (Sohl-Dickstein et al., 2015; Ho et al., 2020; Song et al., 2020b). For example, the reversed (noise→data) Variance Exploding (VE) path has the form

$$p_t(x) = \mathcal{N}(x|x_1, \sigma_{1-t}^2 I), \tag{16}$$

where σ_t is an increasing function, $\sigma_0 = 0$, and $\sigma_1 \gg 1$. Next, equation 16 provides the choices of $\mu_t(x_1) = x_1$ and $\sigma_t(x_1) = \sigma_{1-t}$. Plugging these into equation 15 of Theorem 3 we get

$$u_t(x|x_1) = -\frac{\sigma_{1-t}'}{\sigma_{1-t}}(x - x_1). \tag{17}$$

The reversed (noise→data) Variance Preserving (VP) diffusion path has the form

$$p_t(x|x_1) = \mathcal{N}(x \mid \alpha_{1-t}x_1, \left(1 - \alpha_{1-t}^2\right) I), \text{where } \alpha_t = e^{-\frac{1}{2}T(t)}, T(t) = \int_0^t \beta(s)ds, \tag{18}$$

and β is the noise scale function. Equation 18 provides the choices of $\mu_t(x_1) = \alpha_{1-t}x_1$ and $\sigma_t(x_1) = \sqrt{1 - \alpha_{1-t}^2}$. Plugging these into equation 15 of Theorem 3 we get

$$u_t(x|x_1) = \frac{\alpha_{1-t}'}{1 - \alpha_{1-t}^2}\left(\alpha_{1-t}x - x_1\right) = -\frac{T'(1-t)}{2}\left[\frac{e^{-T(1-t)}x - e^{-\frac{1}{2}T(1-t)}x_1}{1 - e^{-T(1-t)}}\right]. \tag{19}$$

Our construction of the conditional VF $u_t(x|x_1)$ does in fact coincide with the vector field previously used in the deterministic probability flow (Song et al. (2020b), equation 13) when restricted to these conditional diffusion processes; see details in Appendix D. Nevertheless, combining the diffusion conditional VF with the Flow Matching objective offers an attractive training alternative—which we find to be more stable and robust in our experiments—to existing score matching approaches.

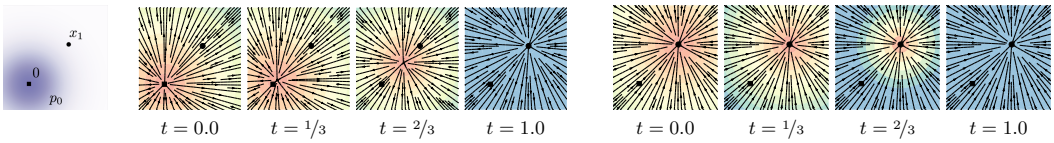
Another important observation is that, as these probability paths were previously derived as solutions of diffusion processes, they do not actually reach a true noise distribution in finite time. In practice, $p_0(x)$ is simply approximated by a suitable Gaussian distribution for sampling and likelihood evaluation. Instead, our construction provides full control over the probability path, and we can just directly set μ_t and σ_t, as we will do next.

Example II: Optimal Transport conditional VFs. An arguably more natural choice for conditional probability paths is to define the mean and the std to simply change linearly in time, *i.e.*,

$$\mu_t(x) = tx_1, \text{ and } \sigma_t(x) = 1 - (1 - \sigma_{\text{min}})t. \tag{20}$$

According to Theorem 3 this path is generated by the VF

$$u_t(x|x_1) = \frac{x_1 - (1 - \sigma_{\text{min}})x}{1 - (1 - \sigma_{\text{min}})t}, \tag{21}$$



Diffusion path – conditional score function OT path – conditional vector field

Figure 2: Compared to the diffusion path's conditional score function, the OT path's conditional vector field has constant direction in time and is arguably simpler to fit with a parametric model. Note the blue color denotes larger magnitude while red color denotes smaller magnitude.

which, in contrast to the diffusion conditional VF (equation 19), is defined for all $t \in [0, 1]$. The conditional flow that corresponds to $u_t(x|x_1)$ is

$$\psi_t(x) = (1 - (1 - \sigma_{\text{min}})t)x + tx_1, \tag{22}$$

and in this case, the CFM loss (see equations 9, 14) takes the form:

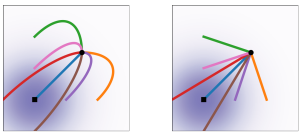
$$\mathcal{L}_{\text{CFM}}(\theta) = \mathbb{E}_{t,q(x_1),p(x_0)} \left\| v_t(\psi_t(x_0)) - \Big(x_1 - (1 - \sigma_{\text{min}})x_0\Big) \right\|^2. \tag{23}$$

Allowing the mean and std to change linearly not only leads to simple and intuitive paths, but it is actually also optimal in the following sense. The conditional flow $\psi_t(x)$ is in fact the Optimal Transport (OT) *displacement map* between the two Gaussians $p_0(x|x_1)$ and $p_1(x|x_1)$. The OT *interpolant*, which is a probability path, is defined to be (see Definition 1.1 in McCann (1997)):

$$p_t = [(1 - t)\text{id} + t\psi]_\star p_0 \tag{24}$$

where $\psi : \mathbb{R}^d \to \mathbb{R}^d$ is the OT map pushing p_0 to p_1, id denotes the identity map, *i.e.*, $\text{id}(x) = x$, and $(1 - t)\text{id} + t\psi$ is called the OT displacement map. Example 1.7 in McCann (1997) shows, that in our case of two Gaussians where the first is a standard one, the OT displacement map takes the form of equation 22.

Intuitively, particles under the OT displacement map always move in straight line trajectories and with constant speed. Figure 3 depicts sampling paths for the diffusion and OT conditional VFs. Interestingly, we find that sampling trajectory from diffusion paths can "overshoot" the final sample, resulting in unnecessary backtracking, whilst the OT paths are guaranteed to stay straight.



Diffusion OT

Figure 3: Diffusion and OT trajectories.

Figure 2 compares the diffusion conditional score function (the regression target in a typical diffusion methods), *i.e.*, $\nabla \log p_t(x|x_1)$ with p_t defined as in equation 18, with the OT conditional VF (equation 21). The start (p_0) and end (p_1) Gaussians are identical in both examples. An interesting observation is that the OT VF has a constant direction in time, which arguably leads to a simpler regression task. This property can also be verified directly from equation 21 as the VF can be written in the form $u_t(x|x_1) = g(t)h(x|x_1)$. Figure 8 in the Appendix shows a visualization of the Diffusion VF. Lastly, we note that although the conditional flow is optimal, this by no means imply that the marginal VF is an optimal transport solution. Nevertheless, we expect the marginal vector field to remain relatively simple.

5 Related Work

Continuous Normalizing Flows were introduced in (Chen et al., 2018) as a continuous-time version of Normalizing Flows (see *e.g.*, Kobyzev et al. (2020); Papamakarios et al. (2021) for an overview). Originally, CNFs are trained with the maximum likelihood objective, but this involves expensive ODE simulations for the forward and backward propagation, resulting in high time complexity due to the sequential nature of ODE simulations. Although some works demonstrated the capability of CNF generative models for image synthesis (Grathwohl et al., 2018), scaling up to very high dimensional images is inherently difficult. A number of works attempted to regularize the ODE to be easier to solve, *e.g.*, using augmentation (Dupont et al., 2019), adding regularization terms (Yang & Karniadakis, 2019; Finlay et al., 2020; Onken et al., 2021; Tong et al., 2020; Kelly et al., 2020), or stochastically sampling the integration interval (Du et al., 2022). These works merely aim to regularize the ODE but do not change the fundamental training algorithm.

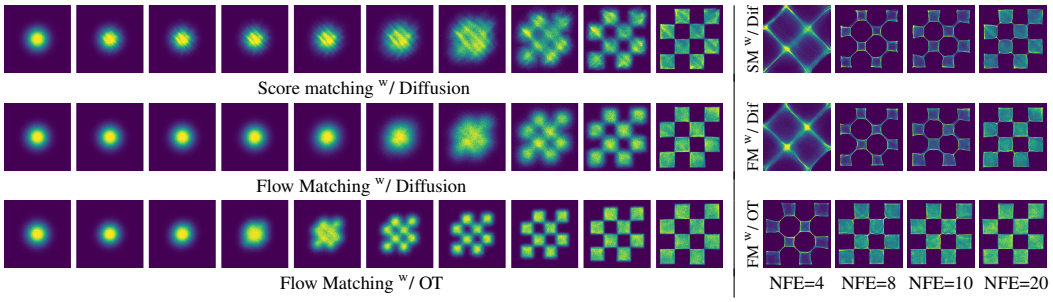


Figure 4: (*left*) Trajectories of CNFs trained with different objectives on 2D checkerboard data. The OT path introduces the checkerboard pattern much earlier, while FM results in more stable training. (*right*) FM with OT results in more efficient sampling, solved using the midpoint scheme.

In order to speed up CNF training, some works have developed simulation-free CNF training frameworks by explicitly designing the target probability path and the dynamics. For instance, Rozen et al. (2021) consider a linear interpolation between the prior and the target density but involves integrals that were difficult to estimate in high dimensions, while Ben-Hamu et al. (2022) consider general probability paths similar to this work but suffers from biased gradients in the stochastic minibatch regime. In contrast, the Flow Matching framework allows simulation-free training with unbiased gradients and readily scales to very high dimensions.

Another approach to simulation-free training relies on the construction of a diffusion process to indirectly define the target probability path (Sohl-Dickstein et al., 2015; Ho et al., 2020; Song & Ermon, 2019). Song et al. (2020b) shows that diffusion models are trained using denoising score matching (Vincent, 2011), a conditional objective that provides unbiased gradients with respect to the score matching objective. Conditional Flow Matching draws inspiration from this result, but generalizes to matching vector fields directly. Due to the ease of scalability, diffusion models have received increased attention, producing a variety of improvements such as loss-rescaling (Song et al., 2021), adding classifier guidance along with architectural improvements (Dhariwal & Nichol, 2021), and learning the noise schedule (Nichol & Dhariwal, 2021; Kingma et al., 2021). However, (Nichol & Dhariwal, 2021) and (Kingma et al., 2021) only consider a restricted setting of Gaussian conditional paths defined by simple diffusion processes with a single parameter—in particular, it does not include our conditional OT path. In an another line of works, (De Bortoli et al., 2021; Wang et al., 2021; Peluchetti, 2021) proposed finite time diffusion constructions via diffusion bridges theory resolving the approximation error incurred by infinite time denoising constructions. While existing works make use of a connection between diffusion processes and continuous normalizing flows with the same probability path (Maoutsa et al., 2020b; Song et al., 2020b; 2021), our work allows us to generalize beyond the class of probability paths modeled by simple diffusion. With our work, it is possible to completely sidestep the diffusion process construction and reason directly with probability paths, while still retaining efficient training and log-likelihood evaluations. Lastly, concurrently to our work (Liu et al., 2022; Albergo & Vanden-Eijnden, 2022) arrived at similar conditional objectives for simulation-free training of CNFs, while Neklyudov et al. (2023) derived an implicit objective when u_t is assumed to be a gradient field.

6 Experiments

We explore the empirical benefits of using Flow Matching on the image datasets of CIFAR-10 (Krizhevsky et al., 2009) and ImageNet at resolutions 32, 64, and 128 (Chrabaszcz et al., 2017; Deng et al., 2009). We also ablate the choice of diffusion path in Flow Matching, particularly between the standard variance preserving diffusion path and the optimal transport path. We discuss how sample generation is improved by directly parameterizing the generating vector field and using the Flow Matching objective. Lastly we show Flow Matching can also be used in the conditional generation setting. Unless otherwise specified, we evaluate likelihood and samples from the model using `dopri5` (Dormand & Prince, 1980) at absolute and relative tolerances of 1e-5. Generated samples can be found in the Appendix, and all implementation details are in Appendix E.